“Meghdoot-AI”

A

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Introduction

Meghdoot.ai is an advanced AI-powered atmospheric nowcasting system designed to accurately predict short-term cloud evolution over a 0–3 hour horizon. The platform utilizes state-of-the-art Latent Diffusion Models (LDM) to overcome the limitations of conventional deterministic forecasting techniques, which frequently generate blurred and physically inconsistent predictions. By learning rich spatio-temporal patterns from high-resolution satellite observations, the system delivers sharper and more realistic cloud movement forecasts.

The solution is built to process indigenous multi-spectral imagery from INSAT-3DR and INSAT-3DS satellites, enabling it to capture complex atmospheric dynamics specific to the Indian subcontinent. Through deep generative modeling, Meghdoot.ai effectively understands cloud formation, dissipation, and motion, providing probabilistic predictions rather than rigid deterministic outputs. This approach ensures more reliable and uncertainty-aware forecasting.

The final product is a scalable, cloud-native dashboard that delivers real-time, high-fidelity visualizations and short-term forecasts. It is specifically tailored to support critical applications such as disaster management, aviation safety, and weather-sensitive operations.

Literature Review

Abstract

Short-term weather forecasting, commonly referred to as nowcasting, is critical for monitoring rapidly evolving atmospheric phenomena such as convective storms, cloudbursts, cyclones, and heavy rainfall events. Accurate prediction of cloud motion within a 0–3 hour horizon remains a challenging task due to the highly non-linear and stochastic nature of cloud dynamics. Traditional nowcasting approaches, including optical flow-based techniques and convolutional recurrent neural networks such as ConvLSTM, often suffer from predictive blur and loss of fine-scale spatial structures. Recent advancements in generative artificial intelligence, particularly diffusion-based models, have demonstrated strong potential in producing sharp, realistic, and probabilistic spatio-temporal predictions. This literature review examines existing research on cloud nowcasting, generative diffusion models, geospatial satellite data processing, and physics-informed artificial intelligence. Special emphasis is placed on the applicability of these methods to indigenous Indian satellite platforms such as INSAT-3DR and INSAT-3DS. The review highlights current limitations, emerging solutions, and identifies a research gap in the development of diffusion-based nowcasting systems optimized for Indian satellite data.

Limitations of Traditional Nowcasting Methods

Early cloud motion prediction techniques were primarily based on optical flow algorithms, which estimate pixel-level displacement between consecutive satellite images (Horn & Schunck, 1981) [3]. While these methods are computationally efficient, they rely on assumptions of smooth and linear motion, making them inadequate for modeling cloud systems that undergo rapid growth,

deformation, or dissipation. Consequently, optical flow methods often fail under convective or cyclonic conditions.

To address these shortcomings, deep learning models such as Convolutional Long Short-Term Memory (ConvLSTM) networks were introduced to capture spatio-temporal dependencies in sequential data (Shi et al., 2015) [2]. Although ConvLSTM-based models demonstrated improved performance over traditional approaches, they are typically trained using pixel-wise loss functions such as mean squared error. This leads to averaged predictions over multiple plausible futures, resulting in the well-known issue of predictive blur (Ravuri et al., 2021) [4]. The loss of fine-scale structural detail significantly limits the operational applicability of these models for real-time disaster response and aviation safety.

Advances in Generative Diffusion Models

Denoising Diffusion Probabilistic Models (DDPMs) introduced a generative framework that models data generation as a gradual denoising process from random noise to realistic samples (Ho et al., 2020) [5]. Unlike deterministic forecasting models, diffusion models learn the full data distribution, allowing them to generate sharp and diverse outputs while naturally capturing uncertainty.

Latent Diffusion Models (LDMs) further improve computational efficiency by operating in a compressed latent space learned using Variational Autoencoders (Rombach et al., 2022) [6]. By reducing the dimensionality of high-resolution inputs, LDMs significantly lower computational cost while preserving essential spatial features. Recent studies such as SEEDS and CorrDiff have demonstrated the effectiveness of diffusion models in weather forecasting and atmospheric downscaling, achieving superior structural fidelity compared to traditional deep learning approaches (Pathak et al., 2022; NVIDIA Earth-2, 2023) [7], [8]. These

findings indicate that diffusion-based models are well-suited for cloud nowcasting tasks.

Geospatial Data Processing for Satellite Imagery

The effective application of deep learning to satellite-based nowcasting requires robust geospatial preprocessing techniques. Satellite imagery is commonly stored in NetCDF or HDF5 formats and includes multi-dimensional information across spatial, spectral, and temporal axes. The xarray library provides efficient tools for handling such n-dimensional datasets, enabling scalable data manipulation and analysis (Hoyer & Hamman, 2017) [9].

Rasterio and GDAL are widely used for georeferencing, reprojection, and resampling satellite imagery onto consistent spatial grids. Proper alignment of infrared (IR), visible (VIS), and water vapor (WV) channels is essential for multi-spectral learning, as misalignment can degrade model performance (Ayzel et al., 2020) [10]. These geospatial data processing techniques form the foundation of reliable satellite-based machine learning pipelines.

Physics-Informed Artificial Intelligence

While generative models excel at producing visually realistic outputs, purely data-driven approaches may generate physically implausible predictions. Physics-informed artificial intelligence seeks to address this limitation by incorporating physical constraints directly into the learning process. Karniadakis et al. (2021) [11] demonstrated that embedding physical laws into neural networks improves generalization and scientific reliability.

Recent research such as Physics-Guided Diffusion for Data Assimilation (PhyDA) integrates physical constraints, including mass conservation, into

diffusion models to ensure physically consistent atmospheric predictions (Zhu et al., 2025) [12]. In the context of cloud nowcasting, such constraints help prevent unrealistic creation or disappearance of cloud mass, thereby increasing trust in AI-generated forecasts for operational use.

Conclusion and Research Gap

The reviewed literature highlights that traditional optical flow and ConvLSTM-based nowcasting methods are fundamentally limited by predictive blur and loss of fine-scale cloud structures. Generative diffusion models, particularly latent diffusion frameworks, offer a promising alternative by producing sharp, realistic, and probabilistic forecasts. However, there remains a significant research gap in the development of diffusion-based cloud nowcasting systems tailored to indigenous Indian satellite platforms such as INSAT-3DR and INSAT-3DS. Addressing this gap is essential for building self-reliant, climate-resilient forecasting tools capable of supporting real-time decision-making in the Indian meteorological context.

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Problem Statement

Traditional short-term weather forecasting models, such as ConvLSTM networks and optical flow–based approaches, struggle to maintain spatial precision over time. These methods are inherently deterministic and rely on pixel-wise regression, which often leads to the phenomenon of “predictive blur.” As forecast horizons increase, fine-grained cloud structures become progressively distorted, resulting in smeared outputs that fail to capture sharp cloud boundaries, rapid convective development, and localized atmospheric changes. This limitation severely reduces their reliability for operational nowcasting applications where accurate depiction of cloud evolution is critical.

In the Indian context, this challenge is further amplified due to the absence of specialized AI-driven forecasting systems tailored to indigenous satellite platforms like INSAT-3DR and INSAT-3DS. While global meteorological models exist, they are not optimized for the unique spatial, spectral, and temporal characteristics of Indian geostationary satellite data. As a result, real-time monitoring and prediction of highly dynamic weather phenomena—such as cloudbursts, thunderstorms, cyclones, and heavy rainfall events—remain inadequate. Existing tools often fail to provide localized, high-resolution forecasts necessary for timely decision-making.

The lack of a self-reliant, AI-based nowcasting framework limits India’s ability to fully leverage its own space assets for rapid weather intelligence. Disaster management agencies, aviation authorities, and urban planners require precise short-term forecasts to mitigate risks and respond effectively to extreme weather events. Without advanced generative models capable of learning complex atmospheric dynamics, current systems remain constrained by low accuracy, poor generalization, and limited scalability.

There is therefore a pressing need for an intelligent, data-driven nowcasting solution that can utilize INSAT satellite imagery effectively, preserve structural cloud details, and generate probabilistic forecasts in real time. Addressing this gap would significantly enhance India's meteorological self-sufficiency and strengthen national capabilities in weather monitoring and disaster preparedness.

Meghdoot.ai implements a two-stage generative pipeline:

1. Compression: A Variational Autoencoder (VAE) compresses high-resolution satellite frames into a latent manifold.

2. Prediction: A U-Net-based Denoising Diffusion model forecasts future states conditioned on historical sequences, ensuring sharp textures and physical consistency through custom loss functions.

Objectives

- To effectively utilize high-resolution multi-spectral imagery from indigenous INSAT-3DR and INSAT-3DS satellites for learning detailed spatio-temporal cloud dynamics, ensuring that the model is optimized for Indian meteorological conditions and data characteristics.

- To implement state-of-the-art latent diffusion models that can generate sharp, high-fidelity, and physically consistent future cloud states, overcoming the limitations of traditional forecasting methods that suffer from predictive blur and loss of structural detail.

- To develop an interactive, cloud-native visualization dashboard that enables real-time interpretation of forecasts, providing intuitive tools for meteorologists, disaster management authorities, and aviation stakeholders to monitor evolving weather patterns and make informed decisions.

- To design and develop an advanced AI-based atmospheric nowcasting system capable of accurately predicting short-term cloud evolution within a 0–3 hour forecast horizon, enabling timely and reliable weather intelligence.

Technical Requirements

For the development of Meghdoot.ai, the technical infrastructure must be robust enough to handle the unique spatio-temporal complexities of Indian meteorological data while maintaining high generative fidelity. The following sections outline the professional-grade hardware and software requirements necessary to support this cloud-native atmospheric nowcasting framework.

Technological Requirements

The project necessitates high-performance computing resources primarily to facilitate the intensive training of the Latent Diffusion Model (LDM) and the initial VAE latent training phases.

Hardware Infrastructure

A GPU-enabled system is mandatory for executing the deep learning workflows, specifically utilizing an NVIDIA T4, V100, or A100 architecture to manage the memory-intensive denoising process and 3-stage fine-tuning. To ensure smooth data manipulation of high-dimensional satellite imagery tensors, a minimum of 16 GB of system RAM is required, though higher capacities are recommended for processing large-scale HDF5 datasets. For infrastructure deployment, the project utilizes scalable cloud environments such as AWS EC2 or Google Colab, which provide the necessary compute elasticity for training complex UNet backbones.

Software Ecosystem

The core development is anchored in Python 3, chosen for its extensive libraries in geospatial science and artificial intelligence. The project utilizes PyTorch and HuggingFace Diffusers as the primary deep learning frameworks to build and train the conditional generative engine. Containerization through Docker is essential for maintaining consistent deployment environments and managing the complex dependencies inherent in meteorological AI pipelines.

Technical Stack

The technical stack is architected to bridge the gap between raw satellite telemetry and actionable weather intelligence through a modular, end-to-end pipeline.

Data Sources and Engineering

The primary data is sourced from indigenous INSAT-3DR and INSAT-3DS satellites, specifically utilizing Level-1C multi-spectral imagery in HDF5 or NetCDF formats. For data engineering, the stack employs xarray to handle n-dimensional labeled arrays, while GDAL and Rasterio are used for critical geospatial tasks such as georeferencing, reprojection, and resampling satellite imagery onto consistent spatial grids. OpenCV and NumPy further support the preprocessing stage by facilitating pixel-level manipulations and normalization for machine learning readiness.

Core Generative Architecture

The system implements a two-stage generative framework. The first stage utilizes a Variational Autoencoder (VAE) to compress high-resolution 512x512 satellite frames into a 64x64 latent space, maximizing computational efficiency. The second stage employs a UNet backbone within a Denoising Diffusion Probabilistic Model (DDPM) to forecast future atmospheric states conditioned on historical sequences.

Loss Functions and Optimization

To ensure scientific validity, the project uses a hybrid loss function combining SSIM (Structural Similarity Index), MAE (Mean Absolute Error), and VGG perceptual loss for visual realism. Crucially, custom physical loss terms, such as Mass Conservation, are integrated to prevent the model from generating scientifically implausible atmospheric dynamics. Weights & Biases is utilized for

real-time experiment tracking and performance monitoring during the training process.

Visualization and Interface

The end product is an interactive, cloud-native dashboard built using Streamlit, which integrates Matplotlib and Plotly to provide high-fidelity animated forecasts. This interface allows meteorologists and disaster management authorities to interpret probabilistic outputs through intuitive, real-time visualizations.

Hardware Requirements

• GPU-enabled system (NVIDIA T4 / V100 / A100)

• Minimum 16 GB RAM

Software Requirements

• Programming Language: Python 3

• Deep Learning Frameworks: PyTorch, HuggingFace Diffusers

• Data Processing: GDAL, Rasterio, xarray, NumPy, OpenCV

• Visualization: Matplotlib, Plotly, Streamlit

• Deployment: Docker, AWS / Google Colab

SWOT Analysis

Strengths

• High prediction accuracy with reduced blur

• Utilization of indigenous satellite data

• Probabilistic forecasting capability

Weaknesses:

• High computational requirements

• Dependency on large satellite datasets

Opportunities:

• Application in disaster management and aviation

• Integration with IMD workflows

Threats:-

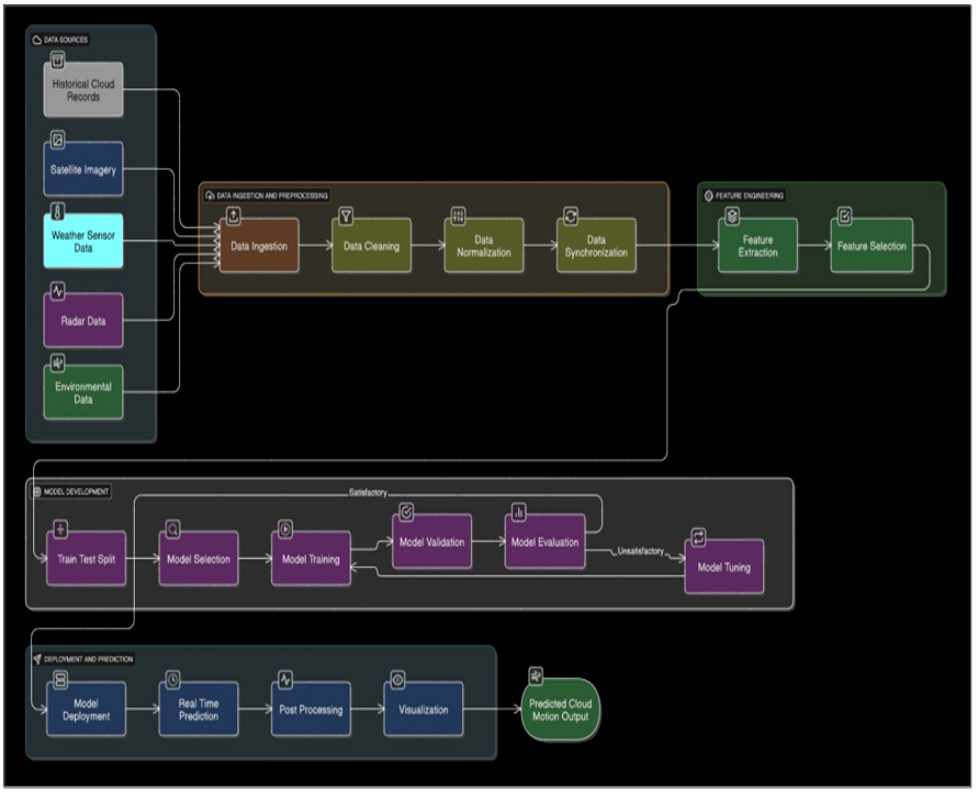
• Data availability constraints

• Rapid evolution of competing AI technologies

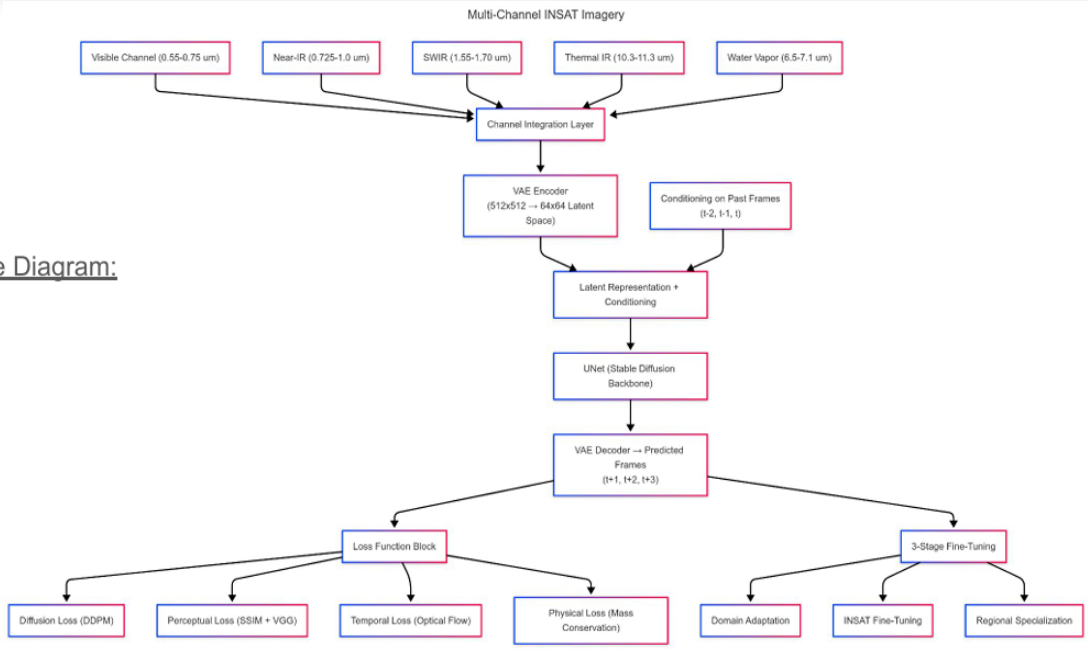
Methodology

The proposed methodology follows a sequential Waterfall model. Initially, INSAT-3DR/3DS Level-1C data is ingested and georeferenced. The processed imagery is compressed using a Variational Autoencoder to obtain latent representations. A diffusion-based UNet model is then trained to predict future latent states conditioned on historical sequences. Physics-informed loss functions are incorporated to ensure mass conservation and temporal consistency. Finally, the predicted outputs are visualized through a Streamlit-based dashboard.

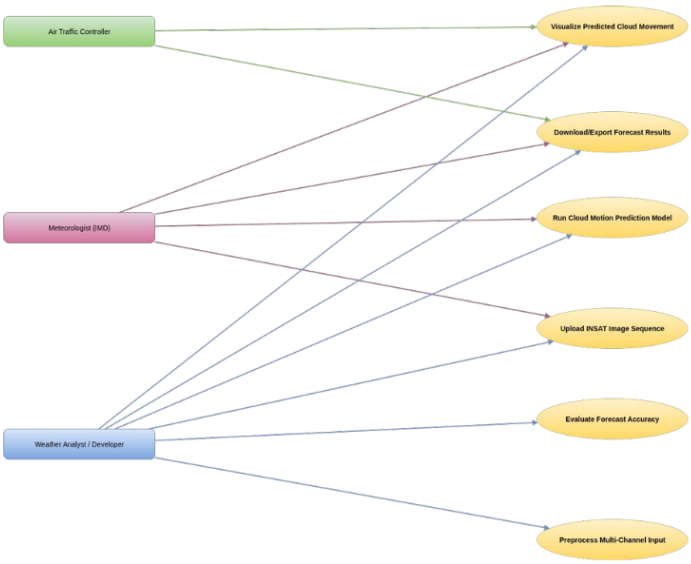
Process Flow Diagram:



Architecture Diagram:



Use Case Diagram:



Architecture Description

Meghdoot-AI is an advanced generative nowcasting framework that integrates geospatial data processing with a two-stage deep learning pipeline to produce high-resolution short-term cloud forecasts. The architecture is designed to efficiently process multi-spectral satellite imagery and generate realistic, temporally consistent predictions of cloud evolution.

Multi-Channel Input Layer

The system begins with the ingestion of multi-spectral satellite data acquired from INSAT-3DR/3DS sensors. The input layer processes diverse spectral bands, including:

Visible (0.55–0.75 μm)

Near Infrared (0.725–1.0 μm)

Short-Wave Infrared (1.55–1.70 μm)

Thermal Infrared (10.3–11.3 μm)

Water Vapor (6.5–7.1 μm)

These channels provide complementary atmospheric information necessary for understanding cloud structure, temperature gradients, and moisture distribution.

Channel Integration and Encoding

The multi-spectral inputs are combined through a dedicated Channel Integration Layer, which aligns and fuses spectral information into a unified representation. This fused data is then passed to a Variational Autoencoder (VAE) Encoder, which performs high-fidelity dimensionality reduction. The encoder compresses a 512×512 spatial input into a compact 64×64 latent space representation, significantly reducing computational complexity while preserving essential cloud features.

Latent Representation and Temporal Conditioning

To capture the temporal evolution of atmospheric patterns, the compressed latent features are conditioned on historical satellite frames from time steps t-2, t-1, and t. This conditioning mechanism enables the model to learn motion trends, cloud dynamics, and contextual dependencies over time, forming a robust foundation for accurate future prediction.

Denoising Diffusion Backbone

The core predictive component of Meghdoot-AI is a U-Net–based Stable Diffusion Backbone. Operating within the learned latent space, this generative engine models future cloud states by iteratively reversing a learned denoising process. Instead of directly predicting pixel values, the diffusion model generates realistic latent representations that maintain structural integrity and fine-grained cloud textures.

Predictive Decoding

The diffusion model produces latent forecasts for future time steps t+1, t+2, and t+3. These predicted latent vectors are passed through the VAE Decoder, which reconstructs them into high-resolution image frames. This decoding process converts abstract latent predictions into visually interpretable and meteorologically meaningful cloud maps.

Scientific Optimization Blocks

To ensure both visual realism and physical plausibility, the model is optimized using a combination of specialized loss functions:

Diffusion Loss (DDPM): Guides the generative process toward accurate denoising.

Perceptual Loss (SSIM + VGG): Enhances structural similarity and visual fidelity.

Temporal Loss (Optical Flow): Enforces motion smoothness across predicted frames.

Physical Loss (Mass Conservation): Ensures consistency with fundamental meteorological principles.

The framework undergoes a structured three-stage fine-tuning process:

- Domain Adaptation to general atmospheric patterns.
- INSAT-Specific Fine-Tuning to optimize for indigenous satellite characteristics
- Regional Specialization tailored to the Indian subcontinent
- This multi-stage refinement ensures that the model is both scientifically robust and operationally reliable.

Process Flow Description

The Meghdoot-AI system follows a structured and modular workflow that transforms raw satellite observations into actionable nowcasts through a sequence of well-defined stages.

Data Sources and Ingestion

The pipeline begins with the collection of heterogeneous meteorological data streams, including:

- INSAT-3DR/3DS Level-1C satellite imagery
- Historical cloud observation records
- Ground-based weather sensor data
- Auxiliary environmental datasets

Data Ingestion and Preprocessing

Raw satellite data undergoes an extensive preprocessing cycle to ensure consistency and quality. Key steps include:

- **Data Cleaning:** Removal of noise, artifacts, and sensor anomalies
- **Data Normalization:** Scaling and standardization of pixel intensities
- **Data Synchronization:** Temporal alignment of multi-spectral channels

Feature Engineering

The preprocessed data is subjected to feature engineering to extract the most relevant atmospheric indicators. This involves:

- **Feature Extraction:** Deriving cloud textures, motion gradients, and spectral patterns
- **Feature Selection:** Identifying the most informative attributes for prediction

Model Development Cycle

The core modeling phase follows an iterative development process with continuous evaluation and refinement:

- **Train-Test Split:** Partitioning datasets for unbiased evaluation
- **Model Selection and Training:** Training the latent diffusion framework
- **Model Validation and Evaluation:** Measuring performance using metrics such as RMSE and SSIM

- **Model Tuning:** Adjusting hyperparameters and loss weights when results are unsatisfactory

Deployment and Prediction

Once validated, the trained model is deployed in a cloud-native environment. The operational phase consists of:

- **Model Deployment:** Hosting on platforms such as AWS or Google Colab
- **Real-Time Prediction:** Generating forecasts as new satellite data becomes available
- **Post-Processing and Visualization**: Enhancing predicted frames and presenting them via an interactive dashboard

The final output is a real-time nowcasting interface that supports stakeholders such as meteorologists, disaster management teams, and air traffic controllers in making informed decisions.

Latent Space Compression

The first stage of the system focuses on efficient representation learning using a Variational Autoencoder (VAE). High-resolution INSAT-3DR/3DS satellite frames contain complex spatial and spectral information that is computationally expensive to process directly. To address this, the VAE learns to compress raw image sequences into a compact latent manifold while preserving essential structural and meteorological features. This dimensionality reduction enables the model to retain meaningful cloud patterns such as texture, motion gradients, and spatial coherence, while significantly lowering computational overhead. The encoder-decoder architecture ensures that the compressed representations remain interpretable and reconstructable, forming a reliable foundation for downstream prediction.

Diffusion-Based Forecasting

In the second stage, future cloud evolution is modeled using a U-Net–based Denoising Diffusion Probabilistic Model (DDPM). The diffusion model operates within the learned latent space, where it forecasts future atmospheric states conditioned on a sequence of historical latent embeddings. Instead of directly predicting pixel values, the model iteratively denoises sampled latent vectors, enabling it to generate sharp, realistic, and temporally coherent cloud structures. This generative approach overcomes the limitations of conventional regression-based models by avoiding error accumulation and predictive blur.

To ensure physical plausibility and meteorological consistency, the system incorporates custom loss functions that balance reconstruction accuracy, temporal smoothness, and structural realism. These losses guide the diffusion process to respect cloud morphology, motion continuity, and spectral characteristics inherent in INSAT data. Additionally, conditioning mechanisms are integrated to capture spatio-temporal dependencies across multiple time steps, allowing the model to learn complex atmospheric dynamics effectively.

End-to-End Inference Pipeline

During inference, recent satellite observations are encoded into latent representations, passed through the diffusion model to generate future latent states, and finally decoded back into image space using the trained VAE decoder. This pipeline enables Meghdoot.ai to produce probabilistic, high-resolution nowcasts over a 0–3 hour horizon with superior visual fidelity and reliability.

Through this two-stage design, Meghdoot.ai combines the efficiency of latent representation learning with the generative power of diffusion models, resulting in a scalable and accurate framework for real-time atmospheric nowcasting.

Project Scope

The scope of Meghdoot.ai encompasses the development of an advanced artificial intelligence–driven nowcasting framework focused on short-term cloud motion prediction over the Indian subcontinent. The system is specifically designed to process and analyze high-resolution satellite imagery from India's indigenous geostationary platforms, INSAT-3DR and INSAT-3DS, to generate accurate, reliable, and real-time forecasts of cloud evolution within a 0–3 hour horizon. By leveraging cutting-edge generative deep learning techniques, Meghdoot.ai aims to bridge the gap between traditional meteorological forecasting systems and modern AI-powered predictive analytics.

At its core, the project focuses on transforming raw multi-spectral satellite data into actionable meteorological insights. The system is intended to function as an end-to-end pipeline, beginning with the ingestion and preprocessing of real-time satellite imagery and culminating in the generation of high-fidelity cloud forecasts. The primary objective is to develop a robust model capable of learning complex spatio-temporal relationships inherent in atmospheric data, thereby enabling precise prediction of cloud movement, formation, and dissipation. This capability is particularly critical for applications that require rapid situational awareness, such as disaster management, aviation operations, and urban planning.

A key aspect of the project scope is its emphasis on real-time nowcasting. Unlike long-range weather prediction models that rely heavily on numerical simulations and atmospheric physics, Meghdoot.ai is tailored for immediate, short-duration forecasting needs. The system is designed to operate with minimal latency, providing continuously updated predictions as new satellite observations become available. This makes it highly suitable for time-sensitive scenarios such as

monitoring developing thunderstorms, tracking cyclonic activity, and predicting localized extreme weather events like cloudbursts and heavy rainfall.

Another major component of the scope is the development of a scalable and adaptable architecture. Meghdoot.ai is envisioned as a cloud-native platform capable of handling large volumes of satellite data efficiently. The system is built to support high-throughput processing, parallel computation, and seamless integration with existing meteorological workflows. This ensures that the solution can be deployed in operational environments such as meteorological departments, emergency response centers, and aviation control systems. Scalability is a central design consideration, enabling the platform to accommodate increasing data loads, additional satellite sources, and expanding user requirements over time.

The project is also scoped to address the unique meteorological and geographical characteristics of the Indian subcontinent. Weather patterns in this region are highly dynamic and diverse, influenced by monsoons, tropical cyclones, and complex land–sea interactions. Global forecasting models are often not optimized for these localized conditions. Meghdoot.ai specifically targets this gap by learning directly from Indian satellite data, ensuring that the predictions are contextually relevant and regionally accurate. The system therefore contributes toward greater meteorological self-reliance by utilizing national space assets to their full potential.

From a technological perspective, the scope includes the implementation of advanced latent diffusion models for cloud forecasting. Traditional deterministic methods often fail to preserve fine structural details in predicted frames, leading to blurred and unreliable outputs. Meghdoot.ai addresses this limitation by employing generative modeling techniques that produce sharper and more realistic predictions. The integration of Variational Autoencoders for latent space compression and diffusion-based generative networks for temporal forecasting

forms a central part of the system's methodological framework. The scope therefore includes research, experimentation, and optimization of deep learning architectures tailored specifically for satellite image sequences.

In addition to predictive modeling, the project scope extends to the development of an interactive visualization and decision-support dashboard. The dashboard is intended to present forecast outputs in an intuitive and user-friendly manner, enabling meteorologists and stakeholders to interpret predictions effectively. Features such as animated cloud motion maps, probabilistic forecast overlays, uncertainty visualization, and region-specific alerts fall within the scope of the system. This interface serves as a critical bridge between complex AI models and real-world operational users, transforming raw predictions into practical insights.

Data preprocessing and management also form an integral part of the project scope. INSAT satellite imagery requires extensive cleaning, calibration, georeferencing, and normalization before it can be used for machine learning applications. The system is therefore designed to include automated pipelines for handling missing data, noise reduction, temporal alignment, and multi-spectral channel processing. Ensuring the quality and consistency of input data is essential for achieving reliable forecasting performance, and as such, data engineering is a major focus area within the scope.

While the current scope is centered on short-term cloud motion prediction, Meghdoot.ai is designed with extensibility in mind. Future enhancements are planned to incorporate multi-spectral data fusion, allowing the model to utilize additional atmospheric variables such as water vapor, infrared channels, and temperature profiles. This would enable more comprehensive weather understanding beyond visible cloud patterns. The framework is also intended to be expandable to longer forecast horizons, improved spatial resolutions, and

integration with auxiliary data sources such as radar observations and numerical weather prediction outputs.

Furthermore, the project scope includes rigorous evaluation and validation of the developed system. Performance metrics such as structural similarity, peak signal-to-noise ratio, and domain-specific meteorological accuracy measures will be employed to assess the quality of predictions. Comparative analysis with existing forecasting approaches will be conducted to demonstrate the advantages of diffusion-based generative models. The system is expected to undergo iterative refinement based on experimental results and real-world testing scenarios.

It is important to note that the scope of Meghdoot.ai is focused exclusively on atmospheric nowcasting and cloud evolution prediction. The project does not aim to replace traditional long-range numerical weather prediction systems, nor does it attempt to forecast unrelated meteorological variables such as temperature, wind speed, or precipitation in its initial implementation. Instead, it is positioned as a complementary AI-driven tool that enhances short-term situational awareness and supports faster decision-making.

Target Audience

Meghdoot.ai is designed as a versatile and high-impact nowcasting platform intended to serve a wide range of stakeholders involved in weather monitoring, disaster preparedness, and atmospheric research. The system is developed to address real-world operational needs where accurate short-term cloud forecasts play a critical role in decision-making. The primary target audience for Meghdoot.ai spans governmental organizations, industry sectors, and academic communities that rely on timely and precise meteorological intelligence.

• Meteorological agencies

One of the primary beneficiaries of Meghdoot.ai is national and regional meteorological agencies responsible for monitoring and forecasting weather conditions across the country. These organizations require continuous situational awareness of evolving atmospheric patterns in order to issue timely forecasts, advisories, and warnings. Traditional numerical weather prediction systems, while effective for long-range forecasting, often lack the temporal resolution and immediacy required for short-term nowcasting applications. Meghdoot.ai fills this gap by providing rapid, AI-driven cloud evolution predictions that enhance the operational capabilities of meteorological departments.

By integrating Meghdoot.ai into their existing workflows, meteorological agencies can improve the accuracy and speed of short-term weather analysis. The system enables forecasters to visualize high-resolution cloud movement, detect rapidly developing weather systems, and assess localized atmospheric disturbances with greater precision. This is particularly valuable during monsoon seasons, cyclonic events, and periods of intense convective activity, where rapid changes in cloud formations can have significant societal impact. The platform

therefore acts as a decision-support tool that strengthens forecasting efficiency and improves the quality of public weather services.

• Disaster management authorities

Disaster management and emergency response organizations form another critical target audience for Meghdoot.ai. Extreme weather events such as cloudbursts, heavy rainfall, cyclones, and thunderstorms often develop with little warning and can lead to severe loss of life and property. In such scenarios, access to accurate short-term weather predictions becomes essential for proactive planning and risk mitigation.

Meghdoot.ai provides disaster management authorities with real-time insights into rapidly changing cloud systems, enabling them to anticipate hazardous conditions well in advance. The ability to predict cloud evolution over the next few hours allows officials to make informed decisions regarding evacuation planning, resource deployment, and public safety measures. For example, early detection of intense cloud formations over vulnerable regions can help authorities issue localized alerts, mobilize rescue teams, and prepare critical infrastructure for potential impact.

The platform’s probabilistic forecasting capabilities further assist disaster response teams in assessing uncertainty and planning for multiple possible scenarios. By offering a clear and intuitive visualization of impending weather threats, Meghdoot.ai supports faster and more effective emergency response, ultimately contributing to reduced disaster risk and improved resilience.

• Aviation sector

The aviation industry is highly sensitive to short-term weather variations, particularly cloud cover, turbulence, and rapidly developing storm systems. Airlines, airport authorities, and air traffic management services depend heavily on accurate and timely meteorological information to ensure safe and efficient flight operations. Meghdoot.ai is specifically tailored to meet the needs of this sector by delivering precise, high-resolution cloud forecasts over short time horizons.

For aviation stakeholders, the system can be used to monitor cloud movement around airports, identify potential storm cells along flight routes, and anticipate visibility changes caused by evolving weather patterns. This information is crucial for flight scheduling, route planning, and minimizing delays or diversions. Ground operations such as takeoff and landing are particularly vulnerable to sudden weather disruptions, and Meghdoot.ai provides an additional layer of situational awareness to support operational safety.

• Researchers in atmospheric science

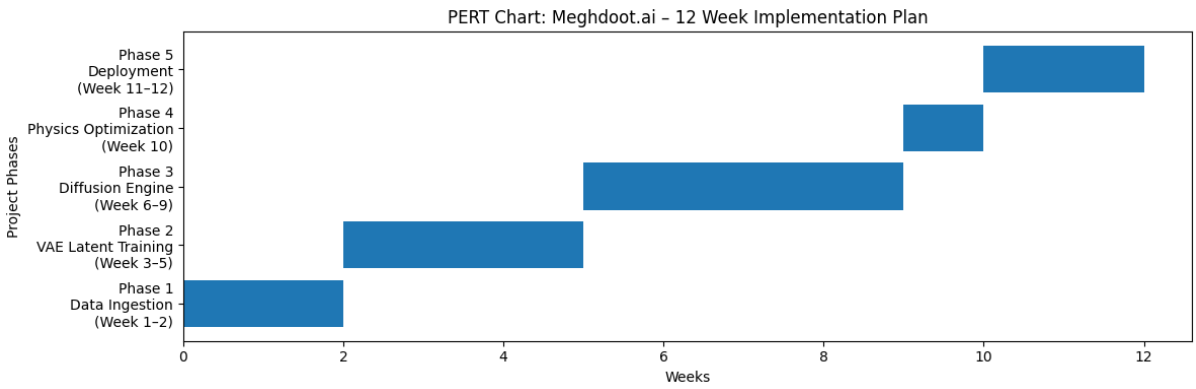
The academic and research community represents another important target audience for Meghdoot.ai. Researchers in atmospheric science, remote sensing, and artificial intelligence require advanced tools to study cloud behavior, weather dynamics, and climate patterns. Meghdoot.ai offers a powerful experimental platform that enables scientists to analyze satellite data in new and innovative ways using modern deep learning techniques.

For researchers, the system provides access to high-quality, AI-generated forecasts that can be used to explore spatio-temporal relationships in cloud formations. The latent diffusion modeling framework implemented in

Meghdoot.ai opens up new possibilities for studying generative approaches to meteorology, allowing academics to investigate alternative forecasting methodologies beyond traditional physics-based models. The platform can serve as a foundation for further research in areas such as extreme weather prediction, data-driven climate modeling, and multi-spectral satellite data analysis.

In addition, Meghdoot.ai can be integrated into educational environments, helping students and early-career researchers gain practical exposure to AI applications in meteorology. By bridging the gap between theoretical research and real-world deployment, the system promotes innovation and contributes to the advancement of atmospheric sciences in India.

Pert Chart



12 - Week Implementation Plan

Phase	Week	Task
Phase 1	Week 1–2	Data Ingestion: Automating the INSAT-3DR Level-1C pipeline and georeferencing.
Phase 2	Week 3–5	Latent Training: Training the VAE to compress 512x512 imagery to 64x64.
Phase 3	Week 6–9	Diffusion Engine: Implementing the UNet backbone for temporal sequence prediction.
Phase 4	Week 10	Physics Optimization: Adding Mass Conservation and Optical Flow loss terms.
Phase 5	Week 11–12	Deployment: Containerizing with Docker and hosting the Streamlit dashboard.

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